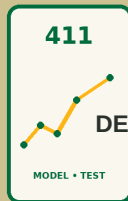
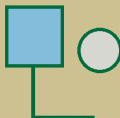
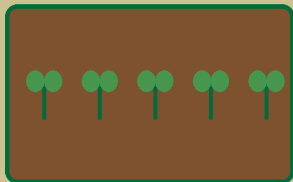


Define a Trial Objective



DEFINE A TRIAL OBJECTIVE

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

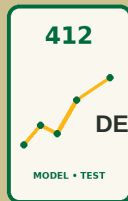
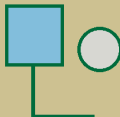
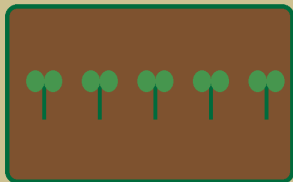
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Develop Treatment Structure



DEVELOP TREATMENT STRUCTURE

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

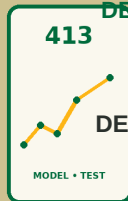
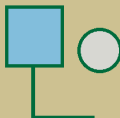
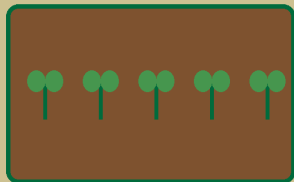
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Design Replication and Randomization



DESIGN REPLICATION AND RANDOMIZATION

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

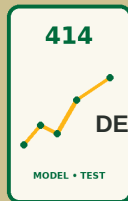
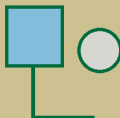
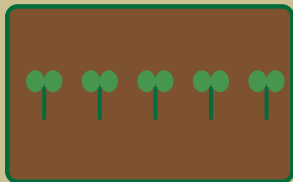
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Create a Block Layout



CREATE A BLOCK LAYOUT

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

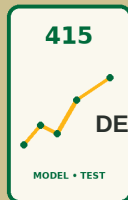
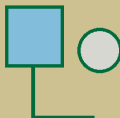
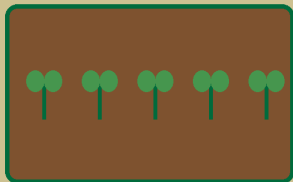
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Define Response Variables



DEFINE RESPONSE VARIABLES

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

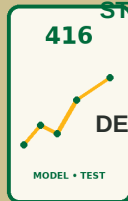
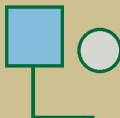
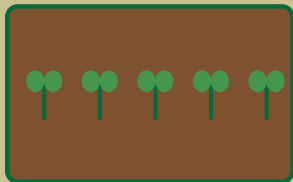
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Standardize a Measurement Protocol



STANDARDIZE A MEASUREMENT PROTOCOL

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

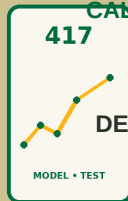
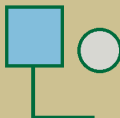
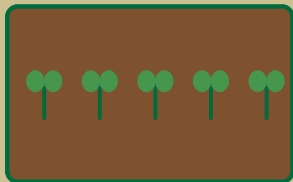
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Calculate Mean and Standard Deviation



CALCULATE MEAN AND STANDARD DEVIATION

DESIGN • INSTRUMENT • MODEL • EVALUATION

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

PA-417 • Calculate Mean and Standard Deviation

TECHNICAL INVESTIGATION

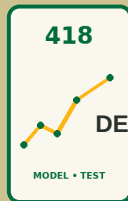
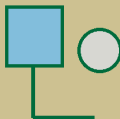
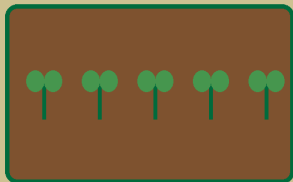
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Interpret a Confidence Interval



INTERPRET A CONFIDENCE INTERVAL

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

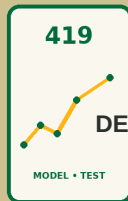
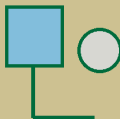
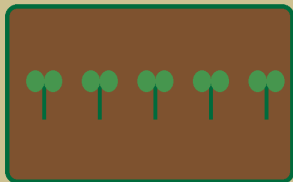
Use quantitative evidence and stated criteria. Compare alternatives or conditions, account for uncertainty, and justify the conclusion.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Estimate Treatment Effect Size



ESTIMATE TREATMENT EFFECT SIZE

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

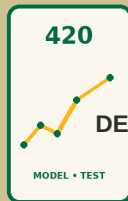
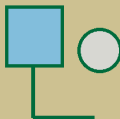
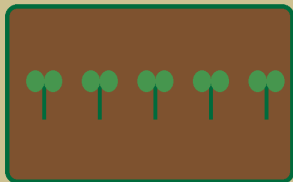
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Write a Trial Recommendation



WRITE A TRIAL RECOMMENDATION

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE